Supplementary Materials for:

Bridging the Gap Between Legacy PCR-based Microsatellite Data with High-Throughput Sequencing Data for Conservation Genomics

Authors:

Dalya Salih[1,2], Ellie E. Armstrong[3,4], Charles T. Robbins[5], Lisette P. Waits[6], Joanna L. Kelley[2,3]

Affiliations:

1. Department of Biomolecular Engineering, University of California, Santa Cruz, 1156 High Street Santa Cruz, CA 95064, USA

2. Department of Ecology and Evolutionary Biology, University of California, Santa Cruz, 130 McAllister Way, Santa Cruz, CA 95060, USA

3. School of Biological Sciences, Washington State University, 431 Heald Hall, Pullman, WA 99164, USA

4. Department of Evolution, Ecology & Organismal Biology, University of California, Riverside, 900 University Ave, Riverside, CA 92521, USA

5. School of the Environment, Washington State University, Pullman, WA, 99164, USA

6. Department of Fish and Wildlife Sciences, College of Natural Resources, University of Idaho, Moscow, ID, 83844, USA

Running title:

Comparing Microsatellite Data in Brown Bears

Email addresses:

dsalih@ucsc.edu, ellie.armstrong@ucr.edu, lwaits@uidaho.edu, ctrobbins@wsu.edu, jokelley@ucsc.edu

Corresponding Author:

Joanna L. Kelley, jokelley@ucsc.edu

Table S1: Microsatellite original publications and primer sequences

Microsat Identity	Original Publication	Forward Primer Sequence	Internal Primer Sequence	Reverse Primer Sequence
G10B	Paetkau et al. 1995	GCCTTTTAATGTTCTGTTGAATTTG	N/A	GACAAATCACAGAAACCTCCATCC
G10C	Paetkau et al. 1995	AAAGCAGAAGGCCTTGATTTCCTG	N/A	GGGGACATAAACACCGAGACAGC
G10H	Paetkau et al. 1998	CAACAAGAAGACCACTGTAA	N/A	AGAGACCACCAAGTAGGATA
G10L	Paetkau et al. 1995	GTACTGATTTAATTCACATTTCCC	N/A	GAAGATACAGAAACCTACCCATGC
G10M	Paetkau et al. 1995	TTCCCCTCATCGTAGGTTGTA	N/A	AATAATTTAAGTGCATCCCAGG
G10P	Paetkau et al. 1995, modified 1998	ATCATAGTTTTACATAGGAGGAAGAAA	N/A	TCATGTGGGGAAATACTCTGAA
G10U	Paetkau et al. 1998	TGCAGTGTCAGTTGTTACCAA	N/A	TATTTCCAATGCCCTAAGTGAT
G10X	Paetkau et al. 1995, modified 1998	CCACCTTCTTCCAATTCTC	N/A	TCAGTTATCTGTGAAATCAAAA
G1A	Paetkau et al. 1995	ACCCTGCATACTCTCCTCTGATG	N/A	GCACTGTCCTTGCGTAGAAGTGAC
G1D	Paetkau et al. 1995	ACAGATCTGTGGGTTTATAGGTTACA	N/A	CTACTCTTCCTACTCTTTAAGAG
Mu11-VIC	Taberlet et al. 1997	AATGTGAAAAAGAAAAGGTAGG	AAGTAATTGGTGAAATGACAGG	GAACCCTTCACCGAAAATC
Mu15	Taberlet et al. 1997	GCCTGACCATCCAACATC	CTGAATTATGCAATTAAACAGC	AAATAAGGGAGGCTTGGGT
Mu23	Taberlet et al. 1997	GCCTGTGTGCTATTTTATCC	AATGGGTTTCTTGTTTAATTAC	TTGCTTGCCTAGACCACC
Mu50	Taberlet et al. 1997	GGAGGCGTTCTTTCAGTTGGT	N/A	TGGAACAAAACTTAACACAAATG
REN145P07	Breen et al. 2001	TGGAAAGGTTTGCACTCTGA	NA	TCCAATTCTTTCTGGGATGC

Table S2: PCR-based microsatellite genotypes

Included as a separate Excel file

Table S3: BLAST results for microsatellite identification in genome

Included as a separate Excel file

Table S4: Microsatellite locations in genome

Microsatellite	Chromosome	Start repeat	End repeat	Motif
G10B	NW_026622786.1	4588098	4588141	TG
G10C	NW_026622863.1	14143647	14143681	AC
G10H	NW_026623053.1	18707046	18707077	AC
G10L	NW_026623111.1	76226159	76226217	AC
G10M	NW_026623111.1	15116048	15116089	AC
G10P	NW_026622797.1	37449572	37449627	AC
G10U	NW_026623078.1	38716579	38716613	GT
G10X	NW_026622763.1	42595152	42595204	TG
G1A	NW_026622963.1	2166851	2166891	TG
G1D	NW_026622797.1	20902220	20902252	CA
Mu11-VIC	NW_026623089.1	83246799	83246842	GT
Mu15	NW_026623019.1	22888534	22888573	TG
Mu23	NW_026622874.1	85857257	85857312	AC
Mu50	NW_026623100.1	50132659	50132713	AC
REN145P07	NW_026622919.1	37502642	37502677	AC

Table S5: Coverage Across Different Depths

Bear	Accession Number	Bear ID	Original Coverage	30x Confirmed Coverage	20x Confirmed Coverage	10x Confirmed Coverage	5x Confirmed Coverage	2x Confirmed Coverage
K	SRX7940190	100973	37	30.5	20.3	10.2	5.1	2.4
L	SRX7940191	100974	36.2	30.5	20.3	10.2	5.1	2.4
P	SRX7940192	100975	55.5	30.5	20.3	10.2	5.2	2.4
A	SRX7940196	100979	53.8	30.1	20.1	10	5.1	2.4
C	SRX7940195	100978	62.8	30.5	20.3	10.2	5.1	2.4
D	SRX7940189	100972	56.7	30.1	20.1	10.1	5.1	2.4
F	SRX7940194	100977	46.3	30.1	20.1	10.1	5.1	2.4
J	SRX7940186	100970	50.2	30.1	20.1	10	5.1	2.4
O	SRX7940187	100971	48	30.5	20.3	10.2	5.1	2.4
W	SRX7940193	100976	46.2	30.4	20.3	10.2	5.1	2.4
Z	SRX7940188	100980	50.5	30.5	20.3	10.2	5.1	2.4

Table S6: Downsampling Coverage Genotype Concordance Across Samples

Microsatellite	30x	20x	10x	5x	2x
G10B	100%	90.91%	63.63%	9.09%	N/A*
G10C	100%	100%	81.82%	63.64%	45.45%
G10H	100%	100%	72.73%	36.36%	27.27%
G10L	100%	100%	100%	63.64%	36.36%
G10M	100%	100%	81.82%	72.73%	N/A*
G10P	100%	100%	90.91%	54.55%	36.36%
G10U	100%	90.91%	100%	72.73%	N/A*
G10X	100%	100%	90.91%	72.73%	N/A*
G1A	100%	100%	100%	54.55%	18.18%
G1D	100%	100%	63.64%	54.55%	N/A*
Mu11	100%	100%	81.82%	36.36%	27.27%
Mu15	90.91%	90.91%	72.73%	36.36%	18.18%
Mu23	100%	72.73%	72.73%	36.36	9.09%
Mu50	100%	100%	90.91%	81.82%	N/A*
REN145P07	100%	100%	81.82%	45.45%	36.36%